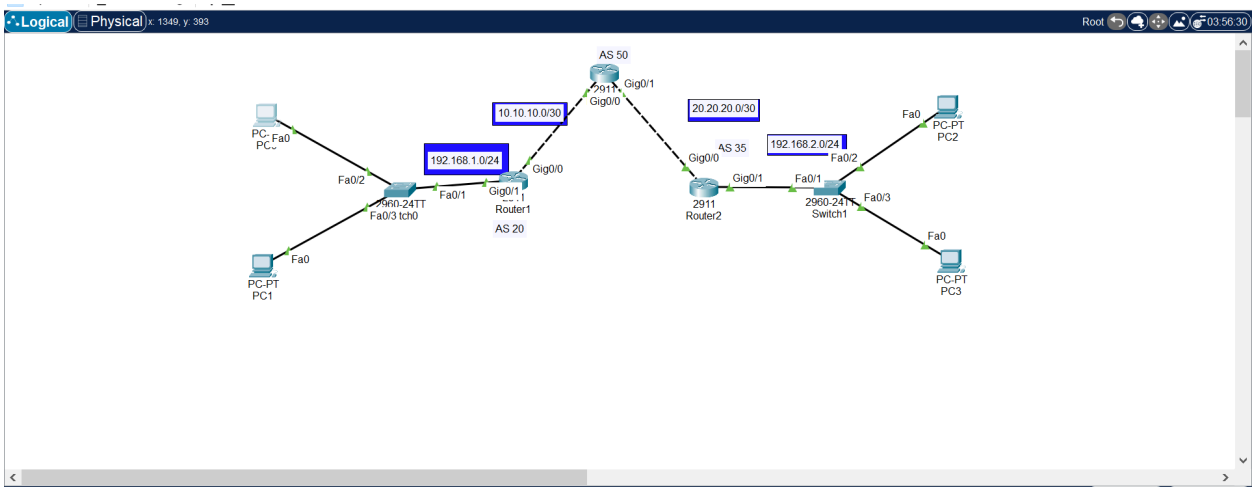
DESIGN AND IMPLEMENTATION OF AN INTER-NETWORK COMMUNICATION SYSTEM USING BORDER GATEWAY PROTOCOL(BGP)

Overview of BGP:

Board Gateway Protocol (BGP) is a dynamic routing protocol used to exchange routing information between different Autonomous Systems(AS). Unlike static routing, BGP automatically learns and advertises networks, making it scalable for interconnecting different networks like your two LANs (192.168.1.0 & 192.168.2.0).

This project has 3 routers, 2 switches, 4PCs to demonstrate BGP communication between two different networks.

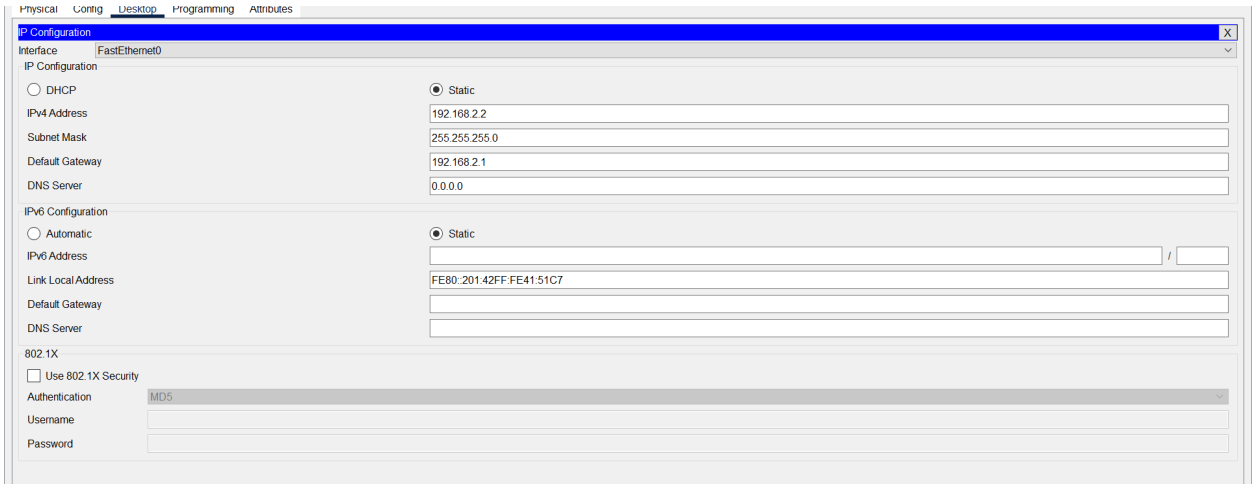


Network Design and IP Addressing scheme:

Device	Ip address	submask
PC0	192.168.1.2	255.255.255.0
PC1	192.168.1.3	255.255.255.0
PC 2	192.168.2.2	255.255.255.0
PC 3	192.168.2.3	255.255.255.0
Router (AS 20)	10.10.10.0	255.255.255.252
Router (AS 50)	10.10.10.0/ 20.20.20.0	255.255.255.252
Router (AS 20)	20.20.20.0	255.255.255.252

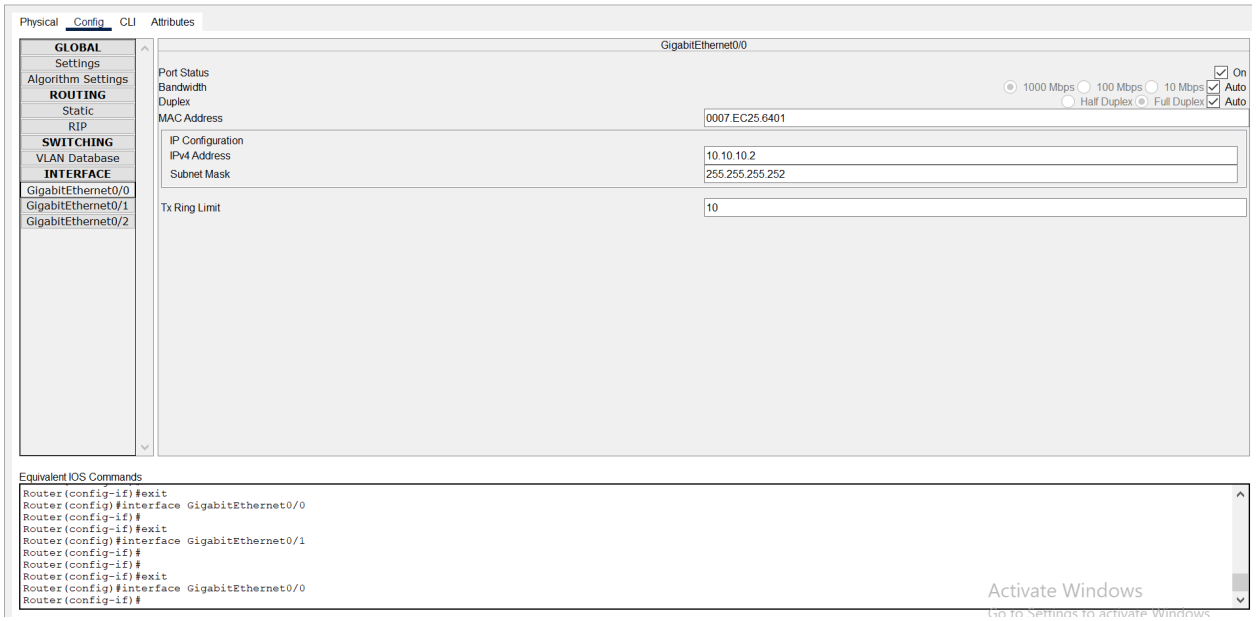
Static IP assignment to PCs:

The 4 PCs were assigned IP addresses statically, meaning IP does not come from DHCP but is typed by the network admin.



Router interface ip assignment :

All routers interfaces were assigned IP addresses using graphics user interface (GUI) > config tab method.



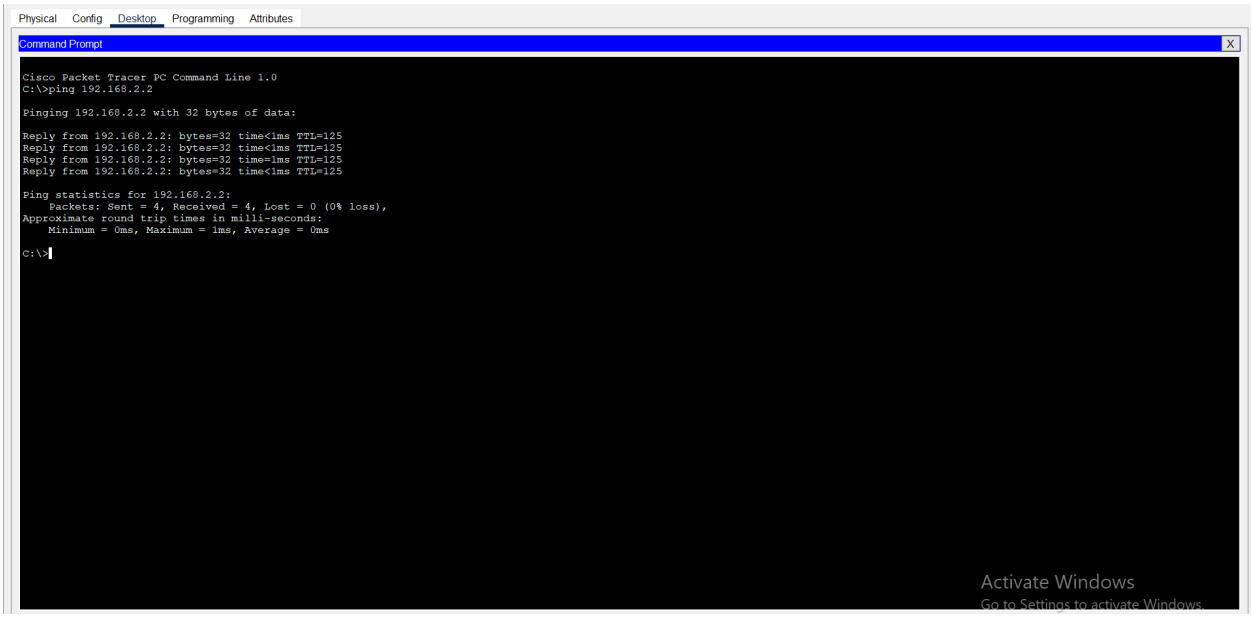
BGP Routing Configuration:

This is the main part. Each router is in a different Autonomous System (AS) to simulate different networks.



Testing and verification:

After configuration, BGP was verified and tested.



Conclusion:

The aim was achieved. BGP successfully advertised 192.168.1.0/24 and 192.168.2.0/24 across 3 Autonomous Systems, allowing PCs in different networks to communicate despite being on different LANs.